

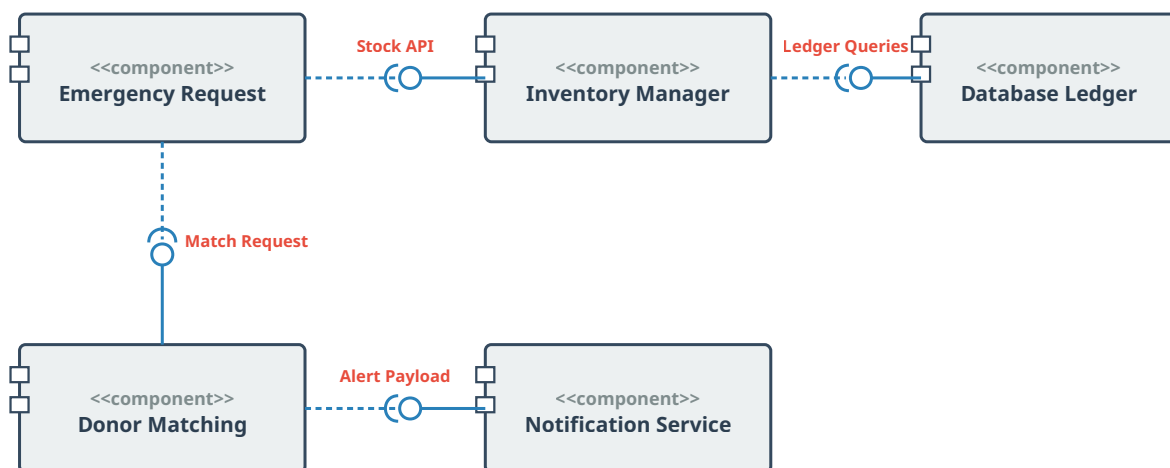
LAB 3: ARCHITECTURE SELECTION & COMPONENT MODELLING

Scenario: Blood Bank Inventory & Emergency Donor Matcher

SRN: PES1UG24CS012

1. UML Component Diagram

The following component diagram illustrates the core modules of the Blood Bank system, depicting at least five distinct components and four explicit interfaces (provided and required) using standard UML ball-and-socket notation.



2. Written Justification

Architectural Choice

We chose the **Microservices Architecture** for the Blood Bank Inventory & Emergency Donor Matcher System.

Specific Reasons for Selection

- **Independent Scalability:** Emergency events inherently create massive, unpredictable spikes in system traffic (e.g., dispatching thousands of geo-targeted SMS alerts instantly to donors within a 10 km radius). A microservices approach allows the Donor Matching and Notification Service components to scale independently during a crisis without bottlenecking the real-time Inventory Manager.
- **Fault Isolation and Reliability:** If the external Notification Service experiences downtime or high latency from a third-party SMS gateway, the failure remains isolated. Hospital staff using the Emergency Request and Inventory Manager components will not be locked out of the system and can continue critical local operations undisturbed.

Security Advantage

Fine-Grained Access Control & Patient-Centric Governance: Microservices enforce strict, bounded contexts. The Donor Matching component can process sensitive demographic and geographical data without exposing that PII (Personally Identifiable Information) to the Emergency Request or Inventory components. This aligns directly with strict healthcare data encryption NFRs and supports advanced, self-sovereign identity models for patient records, ensuring that identity ledgers remain isolated from general hospital stock traffic.

Performance Benefit

Targeted Data Optimization: Different components in this system have vastly different database requirements. A microservices pattern allows the Database Ledger to be heavily optimized for strict ACID properties (transactional consistency to prevent simultaneous blood bag allocation), while the Donor Matching component can utilize specialized, high-speed geospatial indexing to rapidly calculate distance radiuses, preventing infrastructure conflict and reducing overall query latency.